

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

End-to-End Data Analytics Report

Python • MySQL • SQL • Power BI

Prepared from the customer shopping behavior dataset and the Power BI dashboard project

1. Executive Summary

This project analyzes customer shopping behavior using a complete analytics workflow: data preparation in Python, storage in a MySQL database, analytical SQL queries, and interactive visualization in Power BI. The dataset contains 3,900 customer purchase records and 18 attributes covering demographics, products, spending, ratings, discounts, subscriptions, shipping, payment methods, and purchase frequency.

The dataset contains 3,900 records and generates total recorded revenue of \$233,081. The average purchase amount is \$59.76, while the average review rating is 3.75/5. The analysis shows that clothing is the largest revenue category, male customers account for the larger share of recorded revenue, and subscribers do not materially outspend non-subscribers on average.

2. Project Objectives

- Load and inspect customer shopping behavior data using Python and Pandas.
- Create a MySQL database and store the cleaned dataset in a customer table.
- Use SQL to answer ten business-oriented customer behavior questions.
- Build an interactive Power BI dashboard from the MySQL data.
- Translate analytical results into practical business insights and recommendations.

3. Dataset Overview

Metric	Value
Rows / customer records	3,900
Columns	18
Total recorded revenue	\$233,081
Average purchase amount	\$59.76
Average review rating	3.75 / 5
Subscription rate	27.0%
Discount-applied purchase rate	43.0%

Main fields include Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Color, Season, Review Rating, Subscription Status, Shipping Type, Discount Applied, Promo Code Used, Previous Purchases, Payment Method, and Frequency of Purchases.

4. Technology Stack and Architecture

Layer	Technology	Purpose
Data preparation	Python / Pandas	Load, inspect and validate the CSV dataset

Database	MySQL	Store the customer data in database `analysis`
Database access	SQLAlchemy / PyMySQL	Connect Python to MySQL and load the DataFrame
Analysis	MySQL SQL	Answer business questions with aggregations and window functions
Visualization	Power BI Desktop	Create an interactive customer behavior dashboard

Workflow:

CSV → Python/Pandas → SQLAlchemy/PyMySQL → MySQL (analysis.customer) → Power BI → Dashboard

5. Python to MySQL Data Pipeline

The MySQL database used for the project is named `analysis`, and the customer dataset is stored in the `customer` table. The Python connection uses SQLAlchemy with the PyMySQL driver.

```
import pandas as pd
from sqlalchemy import create_engine

df = pd.read_csv("customer_shopping_behavior.csv")

engine = create_engine(
    "mysql+pymysql://<username>:<password>@localhost:3306/analysis"
)

df.to_sql(
    "customer",
    con=engine,
    if_exists="replace",
    index=False
)
```

For security, the database password should not be hard-coded in a portfolio project. Environment variables or a secrets manager are preferable.

6. SQL Analysis and Results

The original SQL contained two important compatibility issues: the dataset uses names such as `Purchase Amount (USD)` rather than `purchase\_amount`, and the expression `::numeric` is PostgreSQL syntax rather than MySQL syntax. The queries below were corrected for MySQL.

Q1. Revenue by Gender

```
SELECT Gender,
       SUM(`Purchase Amount (USD)`) AS revenue
FROM customer
GROUP BY Gender;
```

Gender	Revenue
Male	\$157,890
Female	\$75,191

Insight: Male customers generated \$157,890, compared with \$75,191 from female customers. This is a revenue difference, not proof that male customers have higher individual spending without controlling for customer counts.

Q2. Discount Users Spending Above Average

```
SELECT `Customer ID`, `Purchase Amount (USD)`  
FROM customer  
WHERE `Discount Applied` = 'Yes'  
AND `Purchase Amount (USD)` >= (  
    SELECT AVG(`Purchase Amount (USD)`) FROM customer  
);
```

Result: 839 records satisfy both conditions. The overall average purchase amount is \$59.76.

Q3. Top 5 Products by Average Review Rating

```
SELECT `Item Purchased`,  
    ROUND(AVG(`Review Rating`), 2) AS `Average Product Rating`  
FROM customer  
GROUP BY `Item Purchased`  
ORDER BY AVG(`Review Rating`) DESC  
LIMIT 5;
```

Rank	Product	Average rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

Q4. Standard vs Express Shipping

```
SELECT `Shipping Type`,  
    ROUND(AVG(`Purchase Amount (USD)`), 2) AS `Average Purchase Amount`  
FROM customer  
WHERE `Shipping Type` IN ('Standard', 'Express')  
GROUP BY `Shipping Type`;
```

Shipping type	Average purchase
Express	\$60.48
Standard	\$58.46

Express-shipping customers have a slightly higher average purchase amount than standard-shipping customers in this dataset. This is an association, not evidence that express shipping causes higher spending.

Q5. Subscribers vs Non-Subscribers

```
SELECT `Subscription Status`,  
    COUNT(`Customer ID`) AS total_customers,  
    ROUND(AVG(`Purchase Amount (USD)`), 2) AS avg_spend,  
    ROUND(SUM(`Purchase Amount (USD)`), 2) AS total_revenue  
FROM customer  
GROUP BY `Subscription Status`
```

```
ORDER BY total_revenue DESC;
```

Status	Customers	Avg. spend	Total revenue
No	2,847	\$59.87	\$170,436
Yes	1,053	\$59.49	\$62,645

Insight: subscribers and non-subscribers have almost identical average purchase values (\$59.49 vs \$59.87). Non-subscribers generate more total revenue primarily because they are much more numerous.

Q6. Products with the Highest Discount Rates

```
SELECT `Item Purchased`,
       ROUND(
         100.0 * SUM(CASE WHEN `Discount Applied` = 'Yes' THEN 1 ELSE 0 END) / COUNT(*),
         2
       ) AS discount_rate
FROM customer
GROUP BY `Item Purchased`
ORDER BY discount_rate DESC
LIMIT 5;
```

Rank	Product	Discount rate
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

Q7. Customer Segmentation

```
WITH customer_type AS (
  SELECT `Customer ID`,
         `Previous Purchases`,
         CASE
           WHEN `Previous Purchases` <= 1 THEN 'New'
           WHEN `Previous Purchases` BETWEEN 2 AND 10 THEN 'Returning'
           ELSE 'Loyal'
         END AS customer_segment
  FROM customer
)
SELECT customer_segment,
       COUNT(*) AS `Number of Customers`
FROM customer_type
GROUP BY customer_segment;
```

Segment	Customers
New	83
Returning	701
Loyal	3,116

Important correction: classifying `Previous Purchases = 0` as Loyal would be logically incorrect. The corrected segmentation treats 0–1 previous purchases as New, 2–10 as Returning, and more than 10 as Loyal.

Q8. Top 3 Products within Each Category

```
WITH item_counts AS (  
    SELECT Category,  
           `Item Purchased`,  
           COUNT(`Customer ID`) AS total_orders,  
           ROW_NUMBER() OVER (  
               PARTITION BY Category  
               ORDER BY COUNT(`Customer ID`) DESC  
           ) AS item_rank  
    FROM customer  
    GROUP BY Category, `Item Purchased`  
)  
SELECT item_rank, Category, `Item Purchased`, total_orders  
FROM item_counts  
WHERE item_rank <= 3;
```

Category	Rank	Product	Orders
Accessories	1	Jewelry	171
Accessories	2	Belt	161
Accessories	3	Sunglasses	161
Clothing	1	Blouse	171
Clothing	2	Pants	171
Clothing	3	Shirt	169
Footwear	1	Sandals	160
Footwear	2	Shoes	150
Footwear	3	Sneakers	145
Outerwear	1	Jacket	163
Outerwear	2	Coat	161

Q9. Repeat Buyers and Subscription

```
SELECT `Subscription Status`,  
       COUNT(`Customer ID`) AS repeat_buyers  
FROM customer  
WHERE `Previous Purchases` > 5  
GROUP BY `Subscription Status`;
```

Subscription status	Repeat buyers (>5 previous purchases)	Share
Yes	958	27.6%
No	2518	72.4%

Interpretation: among customers with more than five previous purchases, about 27.6% are subscribers and 72.4% are not. The distribution is very similar to the overall subscription split, so repeat purchasing alone does not appear strongly associated with subscription status in this dataset.

Q10. Revenue Contribution by Age Group

```
SELECT  
    CASE  
        WHEN Age < 25 THEN 'Under 25'  
        WHEN Age BETWEEN 25 AND 34 THEN '25-34'  
        WHEN Age BETWEEN 35 AND 44 THEN '35-44'  
        WHEN Age BETWEEN 45 AND 54 THEN '45-54'  
        ELSE '55+'  
    END
```

```

        END AS age_group,
        SUM(`Purchase Amount (USD)`) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;

```

Age group	Revenue
55+	\$69,590
25-34	\$45,400
45-54	\$45,370
35-44	\$43,463
Under 25	\$29,258

7. Key Business Findings

- Total recorded revenue is \$233,081 across 3,900 records.
- Customers aged 55+ contribute the highest revenue among the defined age groups (\$69,590).
- Clothing is the largest revenue-generating product category, followed by Accessories, Footwear, and Outerwear.
- Subscribers represent 27.0% of records and have an average purchase of \$59.49, virtually the same as non-subscribers at \$59.87.
- Discounting is common, but the products with the highest discount rates are not necessarily the products with the highest revenue.
- Repeat purchasing is widespread, but repeat buyers are not substantially more likely to subscribe based on the >5 previous-purchases threshold.
- Product review averages are tightly clustered; the top-rated products differ only modestly from one another.

8. Power BI Dashboard

The supplied PBIX file contains a one-page report titled 'Customer Behaviour Dashboard'. Its report definition includes KPI cards, a subscription-status donut chart, category-level revenue and sales charts, slicers, and age-related revenue/sales visuals.

Dashboard element	Purpose
KPI cards	Show customer count, average purchase amount, and average review rating
Subscription donut	Compare the percentage of customers by subscription status
Revenue by Category	Identify the strongest revenue-generating product categories
Sales by Category	Compare sales/order volume by category
Slicers	Allow interactive filtering of the dashboard
Revenue by Age Group	Explore revenue across customer age bands
Sales by Age Group	Explore customer/order volume by age

Dashboard quality note: the PBIX visual definitions label two visuals as 'Revenue By Age Group' and 'Sales By Age Group', but the underlying visual field is Age. If the intention is to show the five age bands used in the SQL analysis, a dedicated Age Group calculated column or measure should be used on those visuals. This prevents a mismatch between the chart title and the actual grouping.

9. Connection and Troubleshooting Log

Issue	Cause	Resolution
Unknown database 'analysis'	The connection initially pointed to a database that MySQL could not find.	Verified the database and used `analysis`.
`df` is not defined	The Pandas DataFrame was not present in the active Jupyter kernel.	Load the CSV into `df` before calling `df.to_sql()`; rerun the load cell after a kernel restart.
SQL column errors	SQL used simplified names such as `purchase_amount`, while the CSV contains `Purchase Amount (USD)`.	Use the actual MySQL column names or rename columns consistently.
MySQL syntax error in Q3	The query used PostgreSQL `::numeric` casting.	Removed the PostgreSQL cast and used MySQL-compatible `AVG()` / `ROUND()`.
Microsoft SQL pre-login handshake error	Power BI was using the SQL Server connector rather than the MySQL connector.	Selected the MySQL Database connector.
Missing MySql.Data.MySqlClient	Power BI could not find the MySQL client/provider library.	Install/configure MySQL Connector/NET and restart Power BI.

10. Recommendations

- Standardize database column names for analytics, for example `purchase\_amount`, `customer\_id`, and `subscription\_status`. This reduces quoting and improves SQL readability.
- Use a controlled data-loading script that validates data types and nulls before writing to MySQL.
- Keep database credentials out of notebooks and source control by using environment variables.
- Add a date field in future data collection so revenue and customer behavior can be analyzed over time.
- Compare subscription conversion rates by customer segment, spend band, category, and purchase frequency rather than relying only on overall averages.
- For discount strategy, compare discounted and non-discounted orders by product, margin, and repeat purchase behavior before deciding whether discounts are effective.
- Improve the Power BI dashboard by using explicit Age Group fields, consistent metric naming, informative tooltips, and a small number of decision-focused KPIs.

11. Limitations

- The dataset is a snapshot of customer shopping behavior, so it cannot establish trends over time.
- The analysis is observational; differences between groups should not be interpreted as causal effects.
- Revenue is based on recorded purchase amount and does not include cost, profit margin, returns, or customer acquisition cost.
- The age-group boundaries are analytical choices and can be changed depending on business requirements.
- The segmentation definition is rule-based rather than validated against future customer retention or lifetime value.

12. Conclusion

The project demonstrates a complete practical analytics pipeline from raw customer data to a database-backed business intelligence dashboard. Python and Pandas handle data ingestion, MySQL provides structured storage and SQL-based analysis, and Power BI converts the results into an interactive reporting layer. The most important analytical takeaway is that customer behavior is not explained by a single variable: subscription, discounts, previous purchases, age, category, and shipping all provide different views of purchasing behavior. A

stronger production version would add time-series data, profit/margin fields, customer lifetime value, and validated customer segments.

Appendix A. Recommended MySQL Column Naming

Current column	Recommended analytics name
Customer ID	customer_id
Purchase Amount (USD)	purchase_amount
Item Purchased	item_purchased
Review Rating	review_rating
Subscription Status	subscription_status
Shipping Type	shipping_type
Discount Applied	discount_applied
Previous Purchases	previous_purchases

Source files used: customer_shopping_behavior.csv and the supplied Power BI PBIX report. All numerical findings in this report were calculated directly from the uploaded CSV.